

Folding in $\mathbb{R}$

Origami, Manifolds, and Dimension Reduction

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2026-08-31

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Preface

A rigid origami crease pattern generates a manifold whose true low-dimensional structure is known exactly, in closed form. Rigid folding bends the sheet along creases only; the facets stay flat. The folded surface is therefore isometric to the flat sheet, and geodesic distances measured along the surface are unchanged by folding. The flat crease pattern is not an approximation of the correct two-dimensional embedding. It *is* the correct embedding — the answer key.

That single property is what this book is built on.

Standard benchmarks for dimension reduction — the Swiss roll, the S-curve, the severed sphere — give one fixed geometry and a visual impression of success. A crease pattern gives an exact reconstruction error and a parameterised family of test problems. The fold angle θ runs continuously from a flat sheet to a fully compressed one, so every method’s performance becomes a curve rather than a single number, and no configuration can be cherry-picked.

Chari and Pachter have argued that two-dimensional embeddings of high-dimensional biological data are systematically misleading. The usual rebuttal is anecdotal. On a manifold where the truth is known exactly, the distortion can be *measured* instead of argued about — and, more usefully, the evaluation metrics themselves can be checked against truth. That is the reason this book exists.

What this book is not

It is not a treatment of rigidity theory, computational origami design, or deep generative modelling. It is not a tutorial on `bookdown`, `make`, or project tooling. Where a section could not be tied back to the ground-truth claim above, it was cut rather than kept for completeness.

Impressum

Folding in R: Origami, Manifolds, and Dimension Reduction

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Source repository: <https://github.com/r-heller/folding-in-r>

Online edition: <https://r-heller.github.io/folding-in-r/>

Prose is licensed under CC BY 4.0. Code is licensed under the MIT License; see `LICENSE-CODE.md` in the repository.

This is a living document. The version you are reading was rendered on 2026-08-31.

Acknowledgments

This book stands on two bodies of work that developed independently and rarely cite each other: the mathematics of rigid origami, and the practice of non-linear dimension reduction. Bringing them together is the only original move here; both halves were built by others.

The R package ecosystem this book depends on — `bookdown`, `Rtsne`, `umap`, `igraph`, `vegan`, `torch` and their dependencies — is maintained largely by volunteers. The bibliography in `packages.bib` is generated automatically so that those authors are cited by name rather than thanked in the abstract.

How to use this book

Standing methodological rules

Three rules apply throughout and are not restated in each chapter.

1. **Every stochastic result is reported across at least 20 seeds, and across more where the spread requires it.** t-SNE and UMAP are seed-dependent; a single run is an anecdote. Twenty is a floor, not a design: the count each method needs is measured, by inverting the 0.02 reportable difference against that method's within-cell spread, and a method needing more gets more. `scripts/measure-seed-budget.R` does the measuring and `data/processed/seed-budget.rds` is its record. Seeds are fixed in `_common.R` as `BENCH_SEEDS` and disclosed with every figure.

Where an artefact is produced below the floor — a pilot, or an arm whose cost forbids it — the count is named in the text that reports it, and `scripts/check-artefact-producers.R` will not let it pass unnamed.

2. **Every θ -dependent result is reported as a curve, never as a single number.** Fold angle is a continuous difficulty parameter. Reporting one value invites cherry-picking, and hides exactly the regime changes the benchmark exists to expose.
3. **Every figure carries `fig.alt`.** No result is encoded by colour alone.

What you need

R 4.1 or newer and the packages pinned in `renv.lock`. Run `renv::restore()` once after cloning. There is nothing else to install: the pattern constructors, the sampler and the evaluation metrics are plain scripts in R/, sourced by `_common.R` when the book renders. The code that produces every result in this book is in the same repository as the book, at the same commit.

The Chapter 10 benchmark grid takes hours to compute. It is precomputed and committed under `data/processed/`; `scripts/run-benchmark-grid.R` is kept as the provenance record rather than run during the build.

How the chapters are shaped

Every body chapter uses the same nine sections, in the same order, with the same anchors. The anchors are namespaced by a short chapter mnemonic — `geom-`, `intro-`, `eval-` — because pandoc slugs headings by their text, and twelve chapters each carrying a heading called “Results” would otherwise collide and silently renumber each other’s cross-references.

```
## What this chapter answers  {#<mn>-question}
## Setup                     {#<mn>-setup}
## Background                 {#<mn>-background}
## <free-wording core>       {#<mn>-core}
## Results                   {#<mn>-results}
## Diagnostics               {#<mn>-diagnostics}
## Where it fails            {#<mn>-limits}
## Reproduce this            {#<mn>-reproduce}
## Further reading           {#<mn>-reading}
```

Only the core section is freely worded — it is the one that carries the chapter’s actual argument, and calling it “Core” in every chapter would be worse than useless. Everything else is fixed, and `scripts/lint-chapters.R` fails the build on a chapter that is missing a slot, has them out of order, or anchors one to the wrong namespace.

Read it as a contract with you rather than a template for us. *Where it fails* is in every chapter because every method in this book fails somewhere, and a chapter that could not fill that section would not be finished. *Reproduce this* names the artefact and the script that produced it, so any number in the chapter can be traced to the run that generated it.

Callouts

Five callout classes are defined in `style/style.css`, each with a LaTeX fallback so they survive into the PDF.

- `boxinfo` — context a reader can skip without losing the thread.
- `boximportant` — a result that is easy to misread, or a default that will bite.
- `boxquestion` — something to try, with the answer available in the sources.
- `boxreport` — an aside on provenance, attribution or the state of the literature.
- `boxempty` — an unadorned block, no icon and no colour. It carries the suggested citation in Citing this book.

Write them as knitr block chunks:

```
\begin{boximportant}
umap::umap() defaults to preserve.seed = TRUE.
\end{boximportant}
```

Never as a raw `<div class="boxinfo">` and never as a pandoc fenced div (`:::boxinfo`). Those two forms are not what the stylesheet targets: they render as unstyled markup in HTML and disappear entirely in PDF, where the boxes are

LaTeX environments declared in `style/preamble.tex`. That is how a sibling volume shipped eighty unstyled boxes.

How to read it

Part I builds the object. Part II tests four families of method against it. Part III is the benchmark proper and contains the book’s main contribution — Chapter 9, where the evaluation metrics themselves are checked against exact truth. Part IV asks whether any of it changes a real decision.

Readers who only want the benchmark can start at Chapter 8 and refer back to Chapter 3 for the generative model.

Part I

Folding

Chapter 1

Introduction

Why should a crease pattern tell you anything about a gene expression matrix?

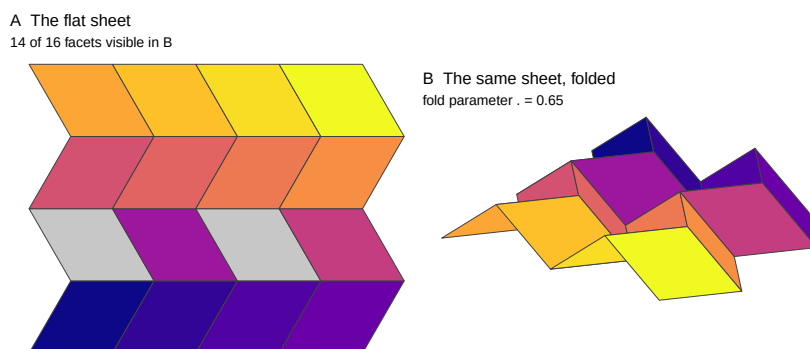


Figure 1.1: A flat crease pattern and the same sheet folded. Turning the folded sheet changes which facets you can see; it does not change the pattern.

The pattern in panel A is the same however panel B is turned. What changes is how much of it any one view exposes — and a dimension-reduction method sees something closer to B than to A. The gap between the two is what the rest of this book measures.

- The ground-truth thesis: a rigid crease pattern *is* the answer key, not a proxy for it.
- What the book claims, and what it explicitly does not claim.
- The Chari & Pachter challenge to 2D embeddings, stated up front as the target.
- Why exact truth lets us evaluate the evaluation metrics, not just the methods.
- Roadmap: fold the sheet (Part I), test the methods (II), build the benchmark (III), change a decision (IV).

Chapter 2

The geometry of folding

What does folding preserve, and what does it destroy?

2.1 What this chapter answers

A sheet of paper folded along creases is a surface in space. Flattened out again it is a region of the plane. The two are the same paper, and that is the whole of this book's method: the flat region is not a good approximation to the folded surface's true two-dimensional structure, it *is* that structure, written down in closed form before any data are generated.

This chapter establishes what that sameness does and does not mean. Folding preserves every distance measured **along** the surface and destroys a great many measured **through** the space around it, and almost every difficulty a dimension-reduction method has with a folded sheet is a consequence of that one sentence. A method sees the second kind of distance and is scored against the first.

Nothing here is asserted. Every claim in this chapter is a check in `tests/testthat/`, and every number is computed while the page is built.

2.2 Setup

```
p <- miura_ori(nx = 6L, ny = 6L, alpha = pi / 3)
f <- fold(p, theta = 0.6)
```

`miura_ori()` returns a flat crease pattern; `fold()` places it in space at a folding parameter θ . Both are in R/, sourced when this book renders, and the notation appendix fixes the symbols.

2.3 Background

2.3.1 What a crease pattern is, and what it is not

A crease pattern is a planar straight-line graph on a region of paper: vertices, straight creases between them, and the facets those creases enclose. It carries one more piece of information per crease, an **assignment** — mountain or valley — which says which way the paper turns there.

The literature on origami mathematics is largely about which patterns fold and how to design one that folds to a given shape (Lang, 2011; Demaine and O’Rourke, 2007). This book needs almost none of that. It needs one family that provably folds, and it needs the folding written as a map it can evaluate at any θ .

Not the Huzita–Hatori axioms. A reader who has met origami mathematics before will expect the construction axioms — the seven operations that say which points and lines a folder can produce with a sequence of folds. They are not what makes a crease pattern a manifold, and they play no part in this book. They describe what can be *constructed*; the question here is what a given pattern, however obtained, *is* once folded. Hull’s *Origametry* [hull2020origametry] covers both and is the place to start if the constructive side is what you came for.

2.3.2 Rigid folding

The foldings this book uses are **rigid**: the facets stay flat and unstretched, and all the motion happens at the creases. That is a strong restriction and it is the source of everything that follows. A folding that let facets bend would still produce a surface, but its flat pattern would no longer be its unfolding, and the answer key would be gone.

Two classical conditions constrain which flat patterns can fold flat at all. At every interior vertex, **Maekawa’s theorem** says the numbers of mountain and valley creases differ by exactly two, and **Kawasaki’s theorem** says the alternating sum of the sector angles around the vertex is zero (Hull, 2020). Both are necessary and neither is sufficient. Chapter 8 returns to how little they guarantee; this chapter uses them as checks rather than as foundations, which is the only honest way to use a necessary condition.

The attribution here is tangled and it is better to say so than to pick a side. Kawasaki’s condition appears independently in work by Justin and by Murata, and no citation resolves cleanly to a single first statement. *Origametry* is the honest reference until one does.

2.3.3 Two kinds of isometry

The word *isometry* does two jobs and the difference between them is this book’s crux.

An **intrinsic** isometry preserves distances measured along the surface — the length of the shortest path an ant could walk without leaving the paper. Rigid

folding is an intrinsic isometry, exactly and by construction: the facets do not stretch, so a path drawn on the flat sheet has the same length after folding.

An **ambient** isometry preserves distances measured through the surrounding space — straight lines through the air. Folding is emphatically *not* an ambient isometry. Bring two points on either side of a crease together by folding, and the straight-line distance between them collapses while the path along the paper does not change at all.

Every method in Part II consumes ambient distance or something built from it. The answer key is intrinsic. That mismatch is not a flaw in the benchmark; it is the benchmark's subject.

2.3.4 Curvature, and where it is not

The natural wrong intuition about a folded sheet is that its curvature has been pushed into the vertices. It is worth writing against rather than merely avoiding, because it is wrong in a way that would undermine the whole construction if it were right.

Discrete Gaussian curvature at a vertex of a polyhedral surface is its angle defect: 2π minus the sum of the sector angles meeting there. For a folded flat sheet those angles are the flat pattern's angles, unchanged by folding, and they sum to exactly 2π because the vertex was an interior point of a flat piece of paper. **The Gaussian curvature is identically zero, at every vertex and on every facet.** Were it otherwise the sheet would not be developable and the flat pattern would not be its unfolding.

What does concentrate on the creases is *extrinsic*: the second fundamental form, which is a measure supported on the one-skeleton rather than a function. The surface is intrinsically flat and extrinsically as bent as one likes.

For manifold learning the consequence that matters is a different quantity entirely. The **reach** of a submanifold (Federer, 1959) is the largest distance within which every point of the ambient space has a unique nearest point on the surface. Sampling theory for manifold learning is generally stated in terms of it (Niyogi et al., 2008), and a crease has reach exactly zero: a point just outside the fold is equidistant from both facets. So the standard sampling guarantees do not apply here at all — not weakly, not approximately. This book's manifolds sit outside the hypotheses of the theory usually invoked to justify the methods being tested on them, which is worth knowing before Part III reports that some of those methods struggle.

2.4 The isometry, in five steps

2.4.1 One: the folding is written down, not solved for

The Miura-ori is a tessellation of congruent parallelograms (Miura, 1985). Its flat vertex (i, j) sits at

$$x = i a + (j \bmod 2) b \cos \alpha, \quad y = j b \sin \alpha,$$

where a and b are the cell side lengths and α its acute angle. The $(j \bmod 2)$ term is the entire pattern: it makes the transverse crease lines zigzag with period two instead of running straight, which is what turns a plain parallelogram lattice into something that folds.

The folded placement is obtained by *positing* a form and solving for the constants rather than by rotating facets about creases and hoping the result closes. Look for

$$x = iL_x + (j \bmod 2) d_x, \quad y = jL_y, \quad z = (i \bmod 2) H.$$

Two facet side vectors follow immediately, $e_1 = (L_x, 0, \pm H)$ for the horizontal edge of flat length a , and $e_2 = (\pm d_x, L_y, 0)$ for the zigzag edge of flat length b . Because both are shared by opposite sides, every facet is a parallelogram and therefore planar with no further conditions. Isometry then needs only the two edge lengths and the angle between them:

$$L_x^2 + H^2 = a^2, \quad d_x^2 + L_y^2 = b^2, \quad L_x d_x = ab \cos \alpha.$$

Three equations in four unknowns: a one-parameter family, which is the rigid folding. Writing $L_x = a \cos \varphi$ and $H = a \sin \varphi$ solves the first identically and leaves $d_x = b \cos \alpha / \cos \varphi$ and $L_y = b \sqrt{1 - \cos^2 \alpha / \cos^2 \varphi}$.

L_y is real only while $\varphi \leq \alpha$, and vanishes exactly at $\varphi = \alpha$, which is the flat-folded state. So φ runs on $[0, \alpha]$ and the natural normalised parameter is $\theta = \varphi / \alpha$.

θ is a parameter, not an angle. It runs on $[0, 1]$: zero is the flat sheet and one is flat-folded. The dihedral angle ρ at a crease is *derived* from it and is never a synonym for it — a distinction this project got wrong once and now states in three places.

2.4.2 Two: the isometry is exact, not converged

Because the placement was derived by imposing edge lengths and facet angles, the isometry holds by construction rather than to a solver's tolerance. That is a claim about algebra, and algebra is exactly the thing that might be wrong, so it is measured.

```
worst_iso <- max(vapply(seq(0, 1, by = 0.05),
                        function(t) facet_isometry_error(p, t), numeric(1)))
```

`facet_isometry_error()` compares **all pairwise distances** between a facet's vertices, folded against flat, not just its edges: an implementation that preserved every edge of a parallelogram while shearing it would pass an edge test and would not be a rigid folding. Over the whole θ sweep the worst departure on this pattern is 0, which is floating-point noise on a closed form rather than a tolerance anyone chose.

Panel A of Figure 2.1 is the answer key. Panel B is what a dimension-reduction method is shown. Everything in Part III is a measurement of the gap between them.

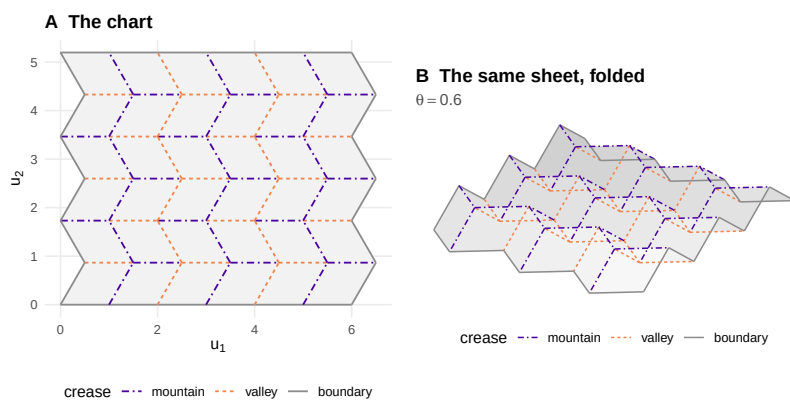


Figure 2.1: The flat crease pattern and the same sheet folded. Mountain and valley creases are distinguished by colour and by line style together, so the distinction survives greyscale printing.

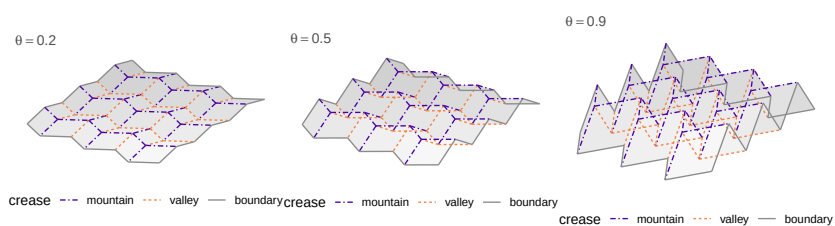


Figure 2.2: One pattern at three points of its one-parameter family. The sheet contracts in both directions as it folds; the flat chart behind it does not change at all.

Figure 2.2 is that family at three values of θ , drawn at one scale. Two things are visible and both matter later. The sheet contracts in **both** directions at once, which is why fold angle is a difficulty parameter rather than a nuisance: it moves the ambient geometry continuously while leaving the chart exactly where it was. And the contraction is not uniform — the far end of the range does most of it, which is why the θ grid in Part III is even in θ and uneven in everything a method cares about.

The derivation above is for one family. Nothing in this book requires a second, and two candidates for one were withdrawn on evidence rather than dropped quietly; Chapter 8 reports both as results. What a reader should take from that is the standard a folding has to meet here: a closed form, an exact isometry, and a fold that reaches every crease family. Two of the three are easy.

2.4.3 Three: what folding does to distance along the surface

Nothing. That is the point, and it needs no further argument once the facets are congruent: a path on the sheet crosses the same facets in the same places at any θ , so its length is the same. Geodesic distance between two sample points is therefore exactly the distance between their positions in the flat chart, subject to one caveat that is stated in `geom-limits` rather than hidden here.

2.4.4 Four: what folding does to distance through space

Here folding does a great deal, and the amount is exact.

Take two points on opposite sides of a single crease, each at perpendicular distance d from it and at the same position along it. Their chart distance is $2d$ at any fold angle. Their ambient distance, once the crease has closed to a dihedral angle ρ , is $2d \sin(\rho/2)$. So

$$\frac{d_A}{d_U} = \sin(\rho/2),$$

which depends on ρ **alone**. Not on d , not on where along the crease the pair sits. That independence is the substance of the claim; a ratio that drifted with distance would mean the relation was an approximation.

```
ratio_err <- max(vapply(seq(pi, 0.05, length.out = 40), function(rho) {
  max(vapply(c(0.1, 1, 7.3), function(d) {
    P <- c( d * sin(rho / 2), 0, d * cos(rho / 2))
    Q <- c(-d * sin(rho / 2), 0, d * cos(rho / 2))
    abs(sqrt(sum((P - Q)^2)) / (2 * d) - sin(rho / 2))
  }, numeric(1)))
}, numeric(1)))
```

Over forty dihedral angles and three separations the worst departure from $\sin(\rho/2)$ is 0 — one unit in the last place, which is what “exact” means in double precision.

A flat sheet has $\rho = \pi$ and contracts nothing. Every fold contracts, and since $\sin(\rho/2) < 1$ for every $\rho < \pi$, **ambient distance is strictly less than chart distance for every $\theta > 0$.**

2.4.5 Five: the floor under “strictly”

That last sentence is false as written, and the chapter says so rather than leaving a reader to trip over it in Part III.

Because $1 - \sin(\rho/2) \approx (\pi - \rho)^2/8$ near the flat state, the contraction is quadratic in the fold angle while machine epsilon is not. There is therefore a fold angle below which the contraction exists mathematically and is not representable.

```
gap      <- function(g) 1 - sin((pi - g) / 2)      # g = pi - rho
floor_pred <- sqrt(8 * .Machine$double.eps / 2)
theta_min <- min(THETA_GRID[THETA_GRID > 0])
```

The predicted floor is 0 radians, and it behaves: `gap()` returns 0 at a fold angle of 10^{-3} and exactly 0 at 10^{-8} . The smallest non-zero θ this book’s grid uses is 0.05, which is above the floor by a wide margin — and that margin is the only reason Part III may treat the contraction as strict.

2.5 Results

Three consequences follow, and they are the reason this chapter comes before the methods rather than after them.

The answer key is not what the methods reproduce. A method that consumes ambient distance is being asked to reproduce a configuration that folding has already contracted. Scoring it against the intrinsic chart is a legitimate question — it is the question of whether the method recovers the *surface* — but it is not the question the method’s loss function asks.

The exact answer scores badly on the wrong metric, provably. This deserves stating as a proposition rather than demonstrating with a simulation.

Proposition. Let stress be measured against ambient distance, and let the candidate embeddings be the exact chart $\mathcal{U}$ and the two-component PCA of the folded cloud. Then for every $\theta > 0$, $\text{stress}(\mathcal{U}) > \text{stress}(\text{PCA})$.

The proof is one line given step four. Ambient-referenced stress measures reproduction of d_A ; the contraction gives $d_A < d_U$ strictly; and two-component PCA is by construction the least-squares-optimal *linear* approximant to that same ambient configuration (Pearson, 1901; Eckart and Young, 1936). The exact unfolding is not competing to reproduce ambient distance and loses to something that is.

An evaluator that ranks the true answer below a plainly wrong one is not malfunctioning. It is answering the question it was asked. Chapter 9 measures how far that goes on real evaluators, and finds it goes further than this proposition requires.

Difficulty has a knob. Because θ is continuous and the contraction is monotone in it, every result in this book is a curve rather than a number. That is standing rule 2, and it is not a stylistic preference. Chapter 8 reports a case where two method families swap places partway along the θ axis, so that the average over θ names a different winner from the curve at seven of its nine points. A benchmark with one fixed geometry cannot produce that observation, because it has nowhere to put it.

This is the specific sense in which a crease pattern is a better instrument than a Swiss roll, and it is worth separating from the claim it is often confused with. It is not that folded sheets resemble real data more closely. It is that the truth is known **and** the difficulty is continuously adjustable, so a method’s performance is a function rather than a point, and the shape of that function is a property of the method. The critique that two-dimensional embeddings distort more than their users admit (Chari and Pachter, 2023; Kobak and Berens, 2019) is an argument this book answers with measurement, and measurement of that kind needs an axis to measure along.

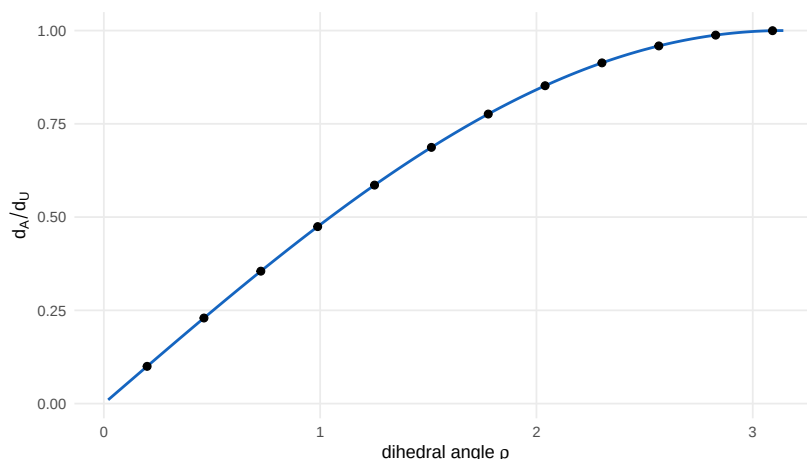


Figure 2.3: The ambient-to-geodesic distance ratio across a single crease, against the dihedral angle. The curve is $\sin(\rho/2)$; the points are measured from folded coordinates. A flat sheet sits at the right-hand end and contracts nothing.

2.6 Diagnostics

The most useful material in this chapter is about two things this project got wrong. Both were caught, both are cheap to state, and both generalise past origami.

2.6.1 An isometry test cannot tell a fold from a pleat

A second pattern family, the Yoshimura, shipped for three days with a folding that was not one. It folded its horizontal creases and left every diagonal at $\rho = \pi$ — an accordion pleat wearing a diamond tessellation’s crease pattern.

It passed the facet-isometry test perfectly, because **a pleat is a rigid folding**. It passed Kawasaki, because the flat pattern was never the problem. Neither test could have caught it: both are properties a correct implementation has, and a pleat has them too.

What caught it was deriving the mountain/valley assignment from the *folded* dihedral angles and testing Maekawa on the result. A pleat leaves whole crease families flat, and a vertex with unfolded creases cannot balance:

```
a <- crease_assignment(p)
iv <- interior_vertices(p)
maekawa <- vapply(iv, function(v) {
  sel <- (p$creases$i == v | p$creases$j == v) & a != "B"
  abs(sum(a[sel] == "M") - sum(a[sel] == "V"))
}, numeric(1))
```

On this Miura all 25 interior vertices satisfy $|M - V| = 2$. On the pleat, twelve of sixteen failed.

2.6.2 A test that a correct implementation passes is not a test a wrong one fails

That is the moral, and the same chapter has to record that this book learned it twice.

The derived assignment above was itself wrong for four days, in a way Maekawa could not see. Maekawa counts labels at a vertex; it is therefore invariant under swapping every mountain for every valley, so it passes on a labelling, on that labelling's exact inverse, and on a third that agrees with neither on half the creases. All three were live in this repository at once, and the one the figures drew was correct on half the interior creases.

```
inverted <- ifelse(a == "M", "V", ifelse(a == "V", "M", "B"))
maekawa_inv <- vapply(iv, function(v) {
  sel <- (p$creases$i == v | p$creases$j == v) & inverted != "B"
  abs(sum(inverted[sel] == "M") - sum(inverted[sel] == "V"))
}, numeric(1))
```

The inverted labelling — every mountain a valley — satisfies Maekawa at 25 of 25 interior vertices, which is all of them. The check that finally settled it computes the assignment a second time by an independent route, from facet normals oriented against the ambient frame rather than against the mesh's own winding, and compares.

2.6.3 A first-order flex at the flat state proves nothing

The third instance, recorded here because it is the same error in a different costume. At $\theta = 0$ every z column of a pattern's rigidity Jacobian vanishes, so the count of non-trivial infinitesimal flexes is $V - 3$ for **any** planar pattern whatsoever — including patterns with no rigid folding at all. A first-order flex at the flat state is a property of being flat, not of being foldable. Second-order

analysis and a finite continuation are what decide, and Chapter 8 records what they decided.

2.7 Where it fails

2.7.1 The chart is the geodesic only inside a convex region

Step three claimed that geodesic distance equals chart distance. That is true when the straight segment between two chart points stays on the paper, and a Miura’s unfolded outline is **not convex** — its edge is a zigzag with teeth. For a pair straddling a tooth the straight chart segment leaves the sheet, the true geodesic must detour, and the chart distance is a *lower bound* rather than an equality.

```
m_in <- sample_manifold(p, theta = 0.6, n = 400L, seed = 1001L)
m_out <- sample_manifold(p, theta = 0.6, n = 400L, seed = 1001L, boundary = TRUE)
exit_in <- chart_exit_fraction(m_in, p)
exit_out <- chart_exit_fraction(m_out, p)
```

`chart_exit_fraction()` measures how often it happens by sampling pairs and walking the segment between them. Drawing strictly inside a convex sub-rectangle gives an exit fraction of 0; including the boundary strip gives 0.013. Every artefact in this book that the answer key is read against is drawn the first way, and the producer that generates the irreducible-loss suite asserts the zero rather than trusting this paragraph.

2.7.2 Self-intersection

Nothing in the algebra of step one prevents a folded placement from passing through itself, and a self-intersecting surface would invalidate every ambient distance measured on it.

```
gap_at <- vapply(c(0.3, 0.6, 0.9), function(t) facet_gap(p, t)$gap, numeric(1))
```

`facet_gap()` computes the exact minimum distance between non-adjacent facets — polygon to polygon, analytically, not by sampling points on them. At θ of 0.3, 0.6 and 0.9 it is 0.851, 0.786 and 0.526: falling as the sheet closes, as it must, and positive throughout.

Figure 2.4 is the whole range. The clearance falls smoothly and stays positive, and the largest θ the benchmark grid uses is marked: at 0.95 the sheet still has 0.396 of clearance. That is the check behind an assumption every ambient distance in this book depends on, and it is a check rather than an argument because the algebra of step one does not forbid self-intersection — it simply does not produce one for this family in this range.

2.7.3 Sampling, and what a finite sample of a zero-reach surface is

The reach remark in `geom-background` has a practical edge that belongs here. Every guarantee about recovering a manifold from a finite sample is stated in

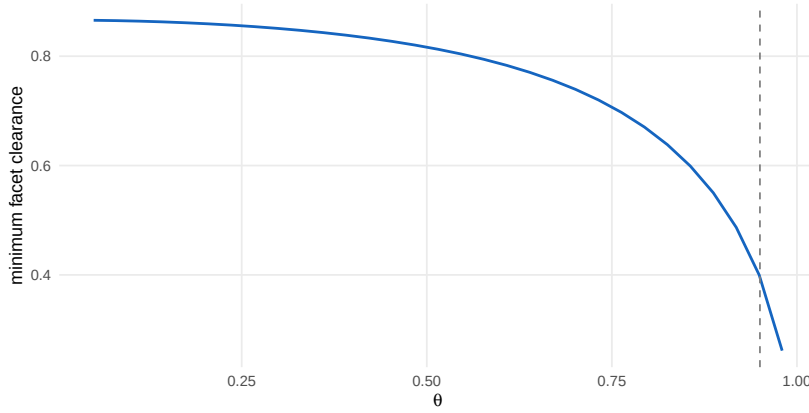


Figure 2.4: Minimum clearance between non-adjacent facets, against the folding parameter. The surface approaches itself as it folds and never touches; the dashed line marks the largest fold angle the benchmark grid uses.

terms of the density of the sample relative to the reach (Niyogi et al., 2008), and a crease has reach zero, so no such guarantee is available at any sample size. What replaces it in this book is not a weaker theorem but a measurement: the sampled surface is checked for how close distant parts of it come, in units of the local spacing, and that ratio is reported alongside every result that depends on it. Isomap’s failure mode on tightly folded surfaces is precisely a short circuit between sheets that the sampling cannot distinguish (Tenenbaum et al., 2000; Balasubramanian and Schwartz, 2002), and the quantity that predicts it is a property of the sample rather than of the surface.

2.7.4 One family

This book ships one pattern family that folds. Two others were specified and both were withdrawn on evidence, which Chapter 8 reports as results rather than as gaps. A benchmark built on a single family cannot support claims about pattern variety, and this book does not make any.

2.7.5 What the chapter does not settle

Whether a folded sheet is a *good* model of the manifolds that arise in real data is not a geometric question and is not answered here. Chapter 11 measures what this benchmark discriminates that the classical ones do not, and Chapter 12 takes the result to a real dataset. The honest position in between is that a crease pattern is a manifold whose truth is known exactly, which is a property no real dataset has and no previously used benchmark provides.

2.8 Reproduce this

Every claim above is an assertion in the test suite, and the suite runs on every push:

- `tests/testthat/test-folding.R` — facet isometry across the θ sweep and the α range, no self-intersection, the dihedral contract.
- `tests/testthat/test-patterns.R` — congruence, Kawasaki, Maekawa, vertex degree.
- `tests/testthat/test-crease-assignment.R` — the mountain/valley derivation against an independent criterion, and the demonstration that Maekawa cannot discriminate.
- `tests/testthat/test-contraction.R` — $\sin(\rho/2)$ and the floating-point floor.

Run them with `Rscript tests/testthat.R` from the repository root.

2.9 Further reading

Hull’s *Origametry* (Hull, 2020) is the single best entry point to the mathematics of flat-foldability and is the reference for both classical theorems above. Demaine and O’Rourke (Demaine and O’Rourke, 2007) is the standard treatment of the computational side, and Lang (Lang, 2011) of design. Miura’s original report (Miura, 1985) introduces the pattern this book uses; Schenk and Guest (Schenk and Guest, 2013) give the mechanics of folded shell structures and are the natural next step for a reader interested in the surfaces rather than the sheets.

On the manifold-learning side, Federer’s reach (Federer, 1959) and its role in sampling guarantees (Niyogi et al., 2008) are what the zero-reach remark above refers to, and Lee and Verleysen (Lee and Verleysen, 2007) is the textbook treatment of the methods Part II takes up.

Chapter 3

Folding as a generative model

How do you turn one pattern into a family of test problems?

- The Miura-ori parameterisation (a, b, α, θ) .
- Uniform sampling on facets, and why naive vertex sampling biases the density.
- Noise models: ambient Gaussian, on-manifold, and outliers.
- What θ does to the k -nearest-neighbour graph, and where short-circuits appear.
- Why every result downstream is a curve over θ rather than a single number.

Part II

Methods under test

Chapter 4

Linear projections and where they break

When is a projection just a shadow?

- PCA via SVD with `prcomp`; why not `princomp` or an eigendecomposition of the covariance.
- At $\theta = 0$ two components are exact. Variance explained collapses as θ rises.
- One curve — and it motivates the whole book.
- Kernel PCA as the bridge into Part II.

Chapter 5

Geodesic methods

If the data lie on a folded sheet, how do you flatten it back out?

- Classical MDS with `cmdscale` as the baseline.
- Isomap: geodesics estimated by shortest paths on a kNN graph (`igraph::distances`).
- Locally linear embedding, and what it assumes about local neighbourhoods.
- Isomap is *built* for isometric embeddings, so the interesting result is where it breaks.
- Report the exact θ at which the kNN graph first bridges two facets — analytically and empirically.

Chapter 6

Neighbour embeddings

Why do t-SNE and UMAP disagree about the same data?

- Rtsne with `check_duplicates = TRUE`; document the duplicate-row failure mode.
- `umap(method = "naive")` — the default backend pulls Python `umap-learn`.
- Perplexity and `n_neighbors` sweeps.
- All results across `BENCH_SEEDS`, reported as distributions rather than means.
- Separate real disagreement from seed noise.

Chapter 7

Learned folds

Can a network learn the fold and the unfold at once?

- A `torch` autoencoder with a two-unit bottleneck. (`keras` would pull TensorFlow and break clean-clone reproducibility.)
- The only method here that learns an explicit inverse: encoder = fold, decoder = unfold.
- Reconstruction loss is the fold–unfold identity made computable.
- Compare the learned decoder against the true unfolding, not just against the input.

Part III

The benchmark

Chapter 8

Building crease-pattern manifolds

How do you build a manifold whose true geometry you already know?

- The helper API in `R/`, end to end.
- Why there is one family and not three. The Yoshimura and the waterbomb were specified here and both were withdrawn: no rigid folding of the Yoshimura is certified in which all three crease families fold, and the waterbomb's is an open question (PLAN.md R1-1 and E2). Both stay in the chapter as documented negative results, which is a stronger section than a list of families would have been – it says what it takes for a pattern to qualify.
- Product constructions: intrinsic dimension $2k$ in ambient $3k$, ground truth still exact.
- Random linear lift to arbitrary ambient dimension D .
- Why a benchmark stuck at intrinsic dimension 2 is not a serious benchmark.

Chapter 9

Ground truth and the evaluators

Your embedding looks good. Is it correct?

- Procrustes-aligned RMSE against the true unfolding as the headline metric.
- Trustworthiness, continuity, kNN preservation, Kruskal stress — each measured *against* truth.
- Where each standard metric is optimistic, and by how much.
- The metrics are kNN-based and mutually correlated; exact truth is what makes the comparison possible.
- The book’s main experimental contribution.

Chapter 10

Benchmark results

Which method survives?

- The full method $\times \theta \times$ noise grid, at least 20 seeds per cell. There is no pattern axis: one family folds, so a pattern factor with a single level is not a factor. Crease count varies instead, which is the knob E1's arm B swept and the one that carries the question.
- Distributions, not means.
- State explicitly how much between-method variation falls inside within-method seed variance.
- Where Isomap wins, where it breaks, and the crossover.
- Grid is precomputed by `scripts/run-benchmark-grid.R` and committed; the script is the provenance record.

Chapter 11

Crease patterns vs. the Swiss roll

Does the new benchmark actually add anything?

- Head-to-head against Swiss roll and S-curve, same methods and same metrics.
- What separates the benchmarks is the SPREAD between methods at matched difficulty, not a flip in their order. E1 looked for a flip and found none; it found a separation, with an interval from a cluster bootstrap over settings and a p-value from cluster-robust regression. A benchmark earns its keep by telling methods apart, and that is the statistic the chapter reports – read from `e1-controlled.rds` through an inline `r` expression, ratio and interval together, never the ratio alone.
- What the crease-pattern family surfaces that a single fixed geometry hides.
- An honest answer to “why not just use the Swiss roll?”

Part IV

Application

Chapter 12

From benchmark to decision

What happens when the data are real?

- Use the Chapter 10 grid to select a method and its hyperparameters.
- Apply that choice to a single-cell dataset.
- Compare against the field-default choice.
- Report a null result honestly if the benchmark-guided choice does not win.
- Closes with visualisation and honest-reporting standards: alt text, no colour-only encoding, seed disclosure, aspect-ratio discipline.

Appendix A

Notation

Symbols

Symbol	Meaning
θ	Folding parameter; the difficulty axis. $\theta = 0$ is the flat sheet
ρ	Dihedral angle at a crease. $\rho = \pi$ when $\theta = 0$; derived from θ , not a second name for it
a, b, α	Miura-ori cell parameters
M, V	Mountain and valley creases
k	Neighbourhood size for kNN-graph methods
D	Ambient dimension after linear lift
$\mathcal{M}$	The folded manifold in $\mathbb{R}^3$
$\mathcal{U}$	The exact unfolding — ground truth in $\mathbb{R}^2$
d_A	Ambient distance, measured in $\mathbb{R}^3$ through the fold
d_U	Intrinsic (path) distance, measured along the surface — equal to Euclidean distance in $\mathcal{U}$

Appendix B

Datasets and codebook

Generated data

All crease-pattern manifolds are generated, not collected. Every dataset in Parts I–III is reproducible from a pattern specification, a fold angle, a sample size, a noise specification, and a seed.

Precomputed results

`data/processed/benchmark-grid.rds` — the Chapter 10 grid. Provenance: `scripts/run-benchmark-grid.R`.

`data/processed/product-grid.rds` — the product suite behind Chapter 8’s irreducible-loss bound, in two arms: a product of two Miura sheets and a product of two Swiss rolls. Provenance: `scripts/run-product-grid.R`.

`data/processed/e1-difficulty.rds`, `e1-controlled.rds`, `e1-armB.rds` — the three arms of E1, which retired the book’s original headline claim. Provenance: `scripts/experiment-e1.R`.

`data/processed/part2-sweeps.rds` — the same cell as the main grid, swept over neighbourhood size instead of over noise. This is where Chapters 4, 5 and 6 answer how much of a method’s result is the method and how much is the k it was handed. Provenance: `scripts/run-part2-sweeps.R`.

`data/processed/classic-grid.rds` — the Swiss roll, the S-curve and the severed sphere, scored by the same function, the same metrics and the same floor as the crease patterns. Chapter 11 needs it: a comparison in which the two sides are scored by two code paths is not a comparison. Provenance: `scripts/run-classic-grid.R`.

`data/processed/evaluator-audit-pilot.rds` — the narrow pilot that decided whether Chapter 9 had a subject. Provenance: `scripts/run-evaluator-audit.R --pilot`.

`data/processed/seed-budget.rds` — the within-cell spread of every method,

and the seed count each one needs before a difference the size of the pre-registered reportable difference (`MIN_REPORTABLE`, fixed in `R/selection.R`) can be detected. This is what standing rule 1's floor is measured against rather than assumed. Provenance: `scripts/measure-seed-budget.R`.

Every one of these carries a provenance block recording the R/ tree hash, the R version, the BLAS, the package versions, the wall clock and whether the tree was dirty when it ran. `read_run()` refuses one that does not, refuses one written by a `--quick` smoke test, and refuses one whose contents are not the contents the chapter around the call was written against.

External data

One dataset, used only in Chapter 12.

Fresh 68k PBMCs (Donor A) — peripheral blood mononuclear cells from a single healthy donor, from Zheng et al. (Zheng et al., 2017). Obtained from the 10x Genomics dataset repository, Cell Ranger 1.1.0 processing, Chromium Universal 3' chemistry.

`scripts/prepare-single-cell.R` downloads the filtered matrix, applies the preprocessing recorded in that script, and writes the artefact Chapter 12 reads. It is not run in CI and not run at render time, on the same provenance convention as the benchmark grid: run it once, commit the result, keep the script as the record of how.

Licence, and what may be redistributed

The dataset is published under **CC BY 4.0**, which permits redistribution with attribution. So the processed matrix *may* ship in this repository, and the constraint on whether it does is size rather than licence — the full filtered matrix is far too large for git, and Chapter 12 does not need it. What is committed is the reduced matrix the chapter actually reads, produced by the script above, with the reduction stated there and in the chapter.

Label provenance, and why the comparison is not circular

This matters more than it looks, and it was an open question until it was checked. Chapter 12's only external evaluation channel is the cell-type labels. If those labels had themselves been derived by clustering one of the embeddings under test, the comparison would be circular and the chapter would be measuring its own assumptions.

They were not. Zheng et al. generated reference profiles by sequencing ten bead-enriched purified subpopulations from the same donor, separately, and assigned each cell in the 68k population the identity of the purified population with which it had the highest Spearman correlation. The labels come from a separate set of experiments, and no clustering or embedding of the matrix under test enters the assignment.

Two caveats belong in the chapter rather than in a footnote, because both bound what a per-label result can mean.

- The assignment is made in gene-expression space — the same space the embeddings are computed from. It is independent of the *embedding*, which is the circularity that would matter here, but it is not an independent measurement of cell identity, and the chapter should not describe it as ground truth. The book reserves that phrase for the crease patterns, where it is earned.
- The authors flag their own assignments as unreliable in specific places: the gap between the highest and second-highest correlation is small for some cells, cytotoxic T against NK being the example they give, and the purified populations are nested — CD4+ T-helper contains all CD4+ cells — so a hand-written rule reassigns cells to the next-highest correlation within a nested family. Any per-label accuracy in Chapter 12 inherits that, and the pre-registered selection rule of Chapter 10 is what keeps it from being read as a ranking.

Glossary

Developable surface A surface that can be flattened into a plane without stretching or tearing.

Dihedral angle (ρ) The angle between the two facets meeting at a crease, measured in the ambient space. A crease in an unfolded sheet has $\rho = \pi$, not 0, and ρ decreases as the sheet folds. It is *derived* from θ and is never a synonym for it.

Folding parameter (θ) The parameter of the folding map, and the difficulty axis of the whole book. $\theta = 0$ is the flat sheet, and the pattern folds monotonically as θ grows. It is a parameter, not an angle read off a figure; the map $\rho(\theta)$ from it to the dihedral angle is stated once per pattern in Chapter 2.

Isometry, intrinsic and ambient An *intrinsic* isometry preserves distances measured **along** the surface. An *ambient* isometry preserves straight-line distance in the space the surface sits in. Rigid folding is the first and emphatically not the second: it contracts ambient distance strictly for every $\theta > 0$. Which of the two a method consumes is the distinction the book turns on, and the reason Chapter 9 reports two headline numbers rather than one.

Mountain / valley assignment The direction of each crease. Written M and V; encoded by linetype as well as colour in every figure.

Reach The largest r such that every point within r of the surface has a unique nearest point on it. Most manifold-learning guarantees are stated in terms of a positive reach; a creased surface has reach exactly zero, which is what puts these objects outside the theory rather than merely at its edge.

Rigid origami Folding in which facets stay flat and all deformation happens at creases.

Short-circuit Two points close in ambient space but far apart geodesically. The failure mode that breaks kNN-graph methods.

Colophon

Written in R Markdown and built with **bookdown** (Xie, 2026). Typeset in Inter and JetBrains Mono. Figures use the viridis plasma palette (option "C"), the series convention for this volume.

Interface icons are Font Awesome Free 6.5.1, vendored in **style/webfonts/** rather than loaded from a content delivery network. The icons are licensed CC BY 4.0, the fonts under the SIL OFL 1.1, and the accompanying code under the MIT License; see <https://fontawesome.com/license/free>.

```
#> - Session info -----
#> setting  value
#> version  R version 4.6.1 (2026-06-24)
#> os       Ubuntu 24.04.4 LTS
#> system   x86_64, linux-gnu
#> ui       X11
#> language (EN)
#> collate  C.UTF-8
#> ctype    C.UTF-8
#> tz       UTC
#> date     2026-08-31
#> pandoc   3.8.3 @ /opt/hostedtoolcache/pandoc/3.8.3/x64/ (via rmarkdown)
#> quarto   NA
#> -----
```


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Citing this book

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Cite the book itself for the argument and the benchmark design. Cite the repository when what you are relying on is the code: the pattern constructors, the folding maps, the metrics, or a committed artefact under `data/processed/`. They carry the same version number, and the repository is the only one of the two that pins a commit.

Heller, R. (2026). *Folding in R: Origami, Manifolds, and Dimension Reduction*. Version 0.1.0. <https://r-heller.github.io/folding-in-r/>

The BibTeX entry is in `citation.bib`, and `CITATION.cff` carries the same metadata in a form GitHub and Zenodo read directly:

```
@book{heller2026folding,  
  title      = {Folding in R: Origami, Manifolds, and Dimension Reduction},  
  author     = {Heller, R.},  
  year       = {2026},  
  publisher  = {Self-published},  
  url        = {https://r-heller.github.io/folding-in-r/},  
  note       = {Version 0.1.0}  
}
```

Both files are generated from the same source and should not be edited by hand; if a version number here disagrees with one of them, the files are right.

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